

Dashboard 4: Loan Repayment & Prepayment Analysis

Introduction:

Understanding **borrower repayment behaviors**, **prepayment tendencies**, and **late payment patterns** is crucial for effective **loan portfolio management**. By examining these dimensions, the dashboard offers actionable insights into **borrower reliability**, **payment discipline**, and **emerging risk trends**. These insights enable financial institutions to **optimize loan structuring**, **identify potential risk factors early**, and **refine credit policies** to enhance overall **portfolio quality** and **minimize default risks**.

DAX Queries:

1. On-Time Repayment Rate (%):

```
On-Time Repayment Rate (%) =  
DIVIDE (  
    CALCULATE (  
        SUM('bondora_data'[nrofscheduledpayments]),  
        'bondora_data'[Delinquency_Category] = "Less than  
30 days",  
        NOT('bondora_data'[Status] IN {"Cancelled"})  
    ),  
    CALCULATE (  
        SUM('bondora_data'[nrofscheduledpayments]),  
        NOT('bondora_data'[Status] IN {"Cancelled"})  
    ),  
    0  
)
```

2. Early Prepayment Rate (%):

```
Early Prepayment Rate (%) =
VAR PaidInFullLoans =
    FILTER(
        'bondora_data',
        'bondora_data'[status] = "Repaid" &&
        NOT(ISBLANK('bondora_data'[lastpaymenton])) &&
        NOT(ISBLANK('bondora_data'[loanduration(months)]))
    )

VAR EarlyPrepaidLoans =
    FILTER(
        PaidInFullLoans,
        DATEDIFF (
            'bondora_data'[loandate],
            'bondora_data'[lastpaymenton],
            MONTH
        ) < 'bondora_data'[loanduration(months)]
    )

RETURN
DIVIDE (
    COUNTROWS(EarlyPrepaidLoans),
    COUNTROWS(PaidInFullLoans),
    0
)

0
)
```

3. Late Payment Rate (%):

```
Late Payment Rate (%) =  
VAR LateLoans =  
    CALCULATE (  
        COUNT ('bondora_data'[loanid]),  
        'bondora_data'[status] = "Late"  
    )  
VAR TotalLoans =  
    COUNT ('bondora_data'[loanid])  
RETURN  
DIVIDE ( LateLoans, TotalLoans, 0)
```

4. Prepayment by Loan Term (%):

```
Prepayment by Loan Term (%) =  
VAR TotalLoans = COUNTROWS (bondora_data)  
VAR PaidLoans = CALCULATE (COUNTROWS  
    (bondora_data), bondora_data[Status] = "Current")  
RETURN DIVIDE (PaidLoans, TotalLoans, 0)
```

5. Average Repayment Period (months):

```
Average Repayment Period (Months) =  
AVERAGEX (  
  
    FILTER (bondora_data, NOT (ISBLANK  
        (bondora_data[LastPaymentOn]))), DATEDIFF  
        (bondora_data[LoanDate], bondora_data[LastPaymentOn], MONTH)  
  
    )
```

Additional Columns:

Repayment Period (Months):

```
Repayment Period (Months) =  
DATEDIFF('bondora_data'[loandate],  
'bondora_data'[maturitydate_last], MONTH)
```

A bin is created using this column for the Histogram

Visualizations:

Visualization	Visualization Type
On-Time Repayment Rate (%)	KPI Card
Early Prepayment Rate (%)	KPI Card
Late Payment Rate (%)	KPI Card
Prepayment by Loan Term (%)	KPI Card
Average Repayment Period (months)	KPI Card
Loan Term vs Prepayment	Bar Chart - X-axis: Loan duration(months) - Y-axis: count (Prepayment by Loan Term (%))
Prepayment Trends Over Time	Line Chart: X-axis: loanapplicationstartdate(Year,Month) Y-axis: Prepayment by Loan Term
Repayment Period Distribution	Histogram: X-axis: Repayment Period(months) (bins) Y-axis: Count(loanid)

Filters & Slicers:

- Loan duration (months)
- Credit Score Grade
- Loan Amount Range
- Loan Status
- Time Range Filter

Conclusion:

Analyzing **borrower repayment behaviors, prepayment patterns, and late payment trends** provides valuable insights into overall **loan performance** and **credit risk**. By tracking these key indicators, the dashboard enhances visibility into **repayment dynamics** and **borrower reliability**. Interactive visualizations enable deeper exploration of **payment timelines, loan terms, and prepayment frequencies**, improving the accuracy of **risk assessment**. Furthermore, the integration of filters and slicers allows stakeholders to examine repayment trends across **borrower segments, loan categories, and time periods**, supporting more **strategic and data-driven lending decisions**.